PLA - 1004

LATERAL THINKING.

TEE Prep.

Date
Page No.

(1) Cubes :

$(24)^3$ $\quad 2\times4\times3 = 24$

$\underline{(2)^3}$ $\quad \underline{24\times2}$ $\quad \underline{24\times4}$ $\quad \underline{(4)^3}$

$\boxed{000}$ $\quad \boxed{00}$ $\quad \boxed{0}$

8000
4800
960
64

13824 ⇒ $\boxed{13824}$

Final Answer.

16×4
64

$(72)^3$ $\quad 7\times2\times3 = 42$

$\underline{(7)^3}$ $\quad \underline{42\times7}$ $\quad \underline{42\times2}$ $\quad \underline{(2)^3}$

$\boxed{000}$ $\quad \boxed{00}$ $\quad \boxed{0}$

483000
~~~~
50400
1400
8
534848

⇒ ~~534848~~

Ans: 373248

$49\times7$
483
$72\times7$
504
$72\times2$
144
$14\times3$
42

⇒ $(125)^3$ ⇒ $(12)\,8\,(5)$ $\quad 12\times5\times3$

$\underline{(12)^3}$ $\quad \underline{180\times12}$ $\quad \underline{180\times5}$ $\quad \underline{(5)^3}$

000 $\quad$ 00 $\quad$ 0

$180\times12$
360
180
2160

$\boxed{\text{Ans}= 1953125}$

Teacher's Signature :
~~~~

② Data Interpretation (Refer PPT)

① Manufactured | 2009 − Manufactured | 2008 =

$\Rightarrow$ ~~52.5~~ ~~53 − 47~~

52.5 − 47.5 = ~~5~~ 5.0

= 5000.

45
46
47
48
49
50
51
52
53
54
55

① ~~no. of products~~:

sold = 25
Manufactures = 35

$\frac{\cancel{72} \times 7}{\cancel{504}}$

$\frac{\cancel{25}\,5}{\cancel{35}\,7} \times 100$

$\frac{500}{7}$

71 / 500
497
3

497

① previous year: ~~x x x~~

$\boxed{\% \text{ increase} = \frac{\text{New val} - \text{Old val}}{\text{Old v}} \times 100}$

①

$\frac{\cancel{78}\,39}{\cancel{128}\,\cancel{64}\,32\,16} \times \cancel{100}\,\cancel{50}\,25$

$\cancel{39} \times 6 \times 9$

39 × 25
195
78
975

16) 975 (6.9
96
150
144

3
× 16 × 6
96

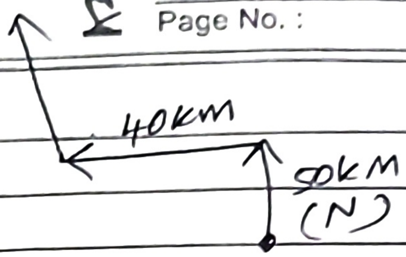
16 24

9500 9520 92500

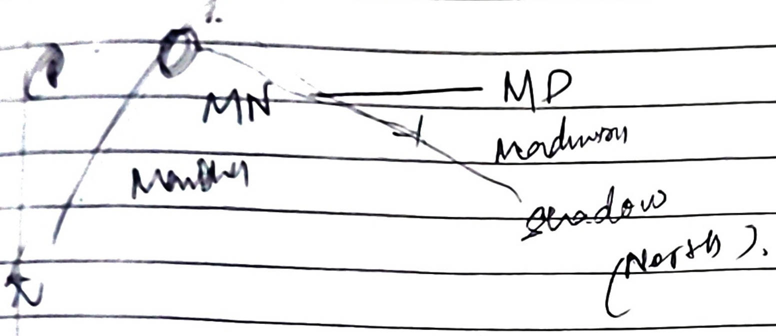
Expt No. 3

Directions: (Refer PPT)

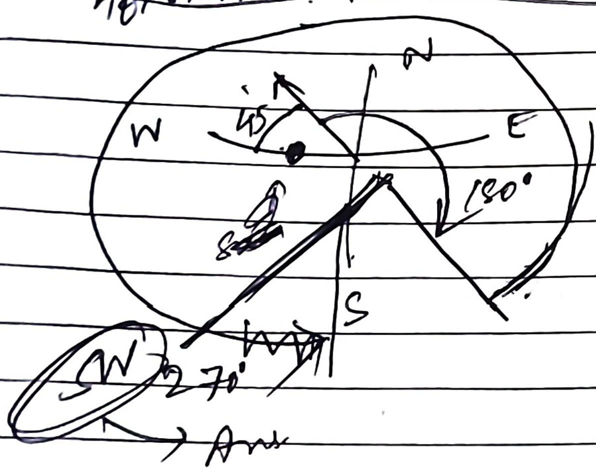
Q

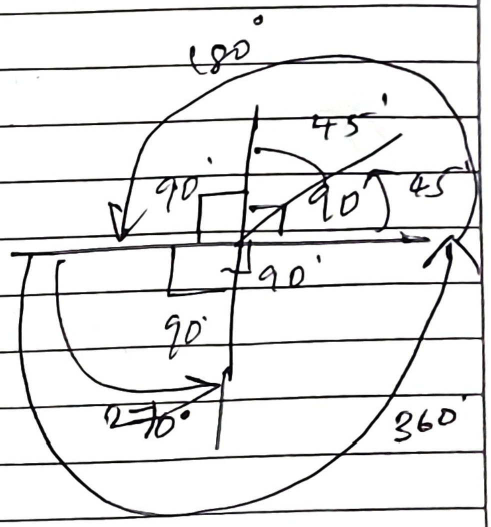


Q

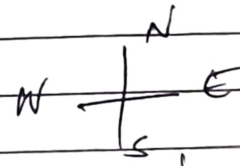


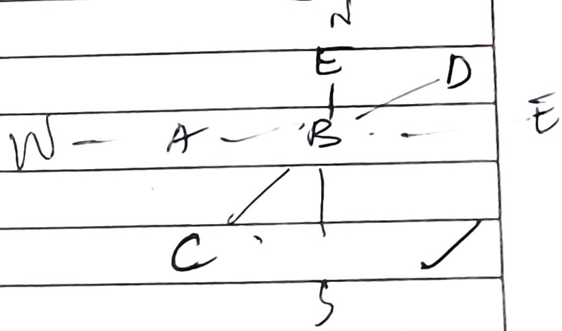
Q ~~Afternoon~~ ~~Sun~~ is W.





Q 5 Persons A B C D E.





* * *

(IX) Data Sufficiency :

1

Mother

25 years

35 years

Rahul

(Ans: 1974)

Brother (1964)

$\cancel{\frac{x}{4} = 5\,kg}$ ~~weight of pole~~

$\cancel{x = 20\,kg}$ $\cancel{2x}$

weight on one pole: x

$$\frac{x}{4} = 5\,kg$$

$$x = 20\,kg$$ — 1 pole.

$\cancel{3x = 20\,kg}$

$\cancel{3x}$ - total weight

$3x = 20$

15 P

P —— Q

1 2 3 ... 15

Strategy (Refer one Note)

If there are n variables, then we need n equations to tell that this is sufficient.

2 variables → 2 E

⇒ Count no. of variables & no. of Equations given in question.

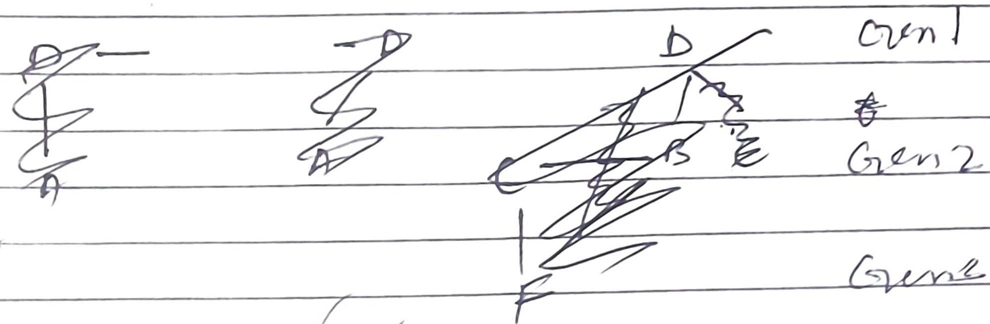
to solve.

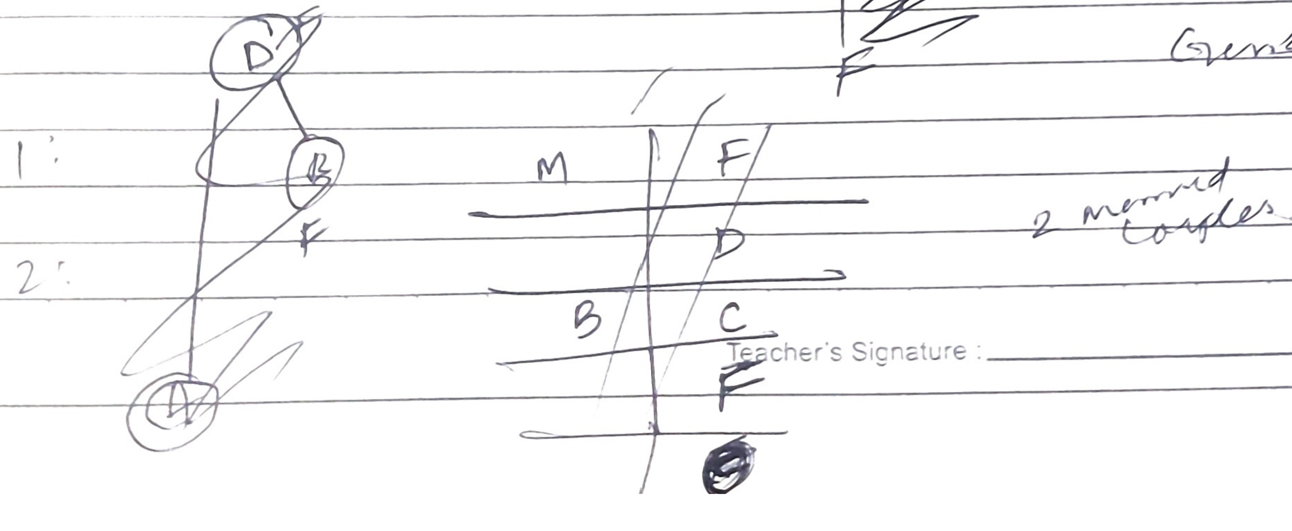
2
4
6
8

5 → **Blood Relations:**

6 members

① A B C D E F.

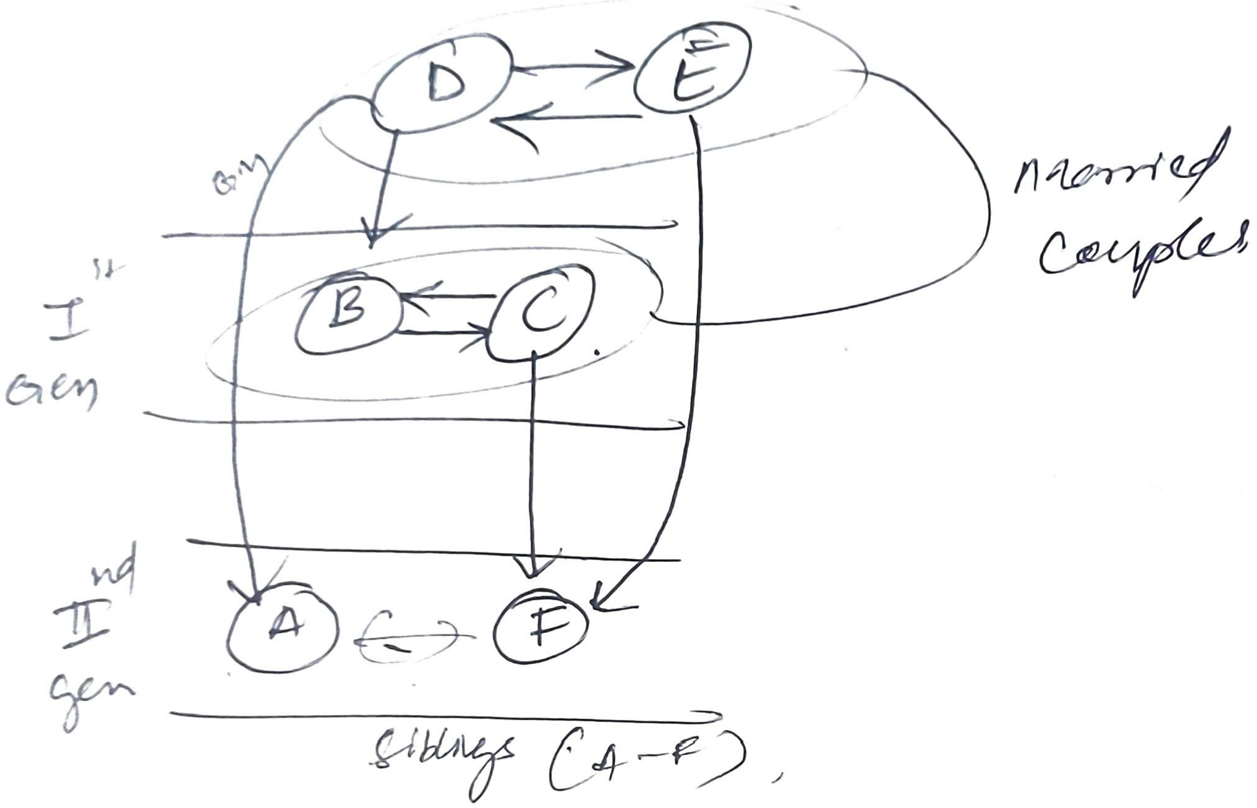




FAMILY

6 members ⇒ A, B, C, D, E, F

2 Married Couples.



Newphew → Your sibling's Son

Niece → Your sibling's Daughter

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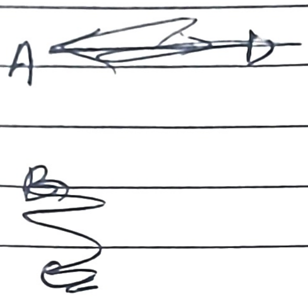
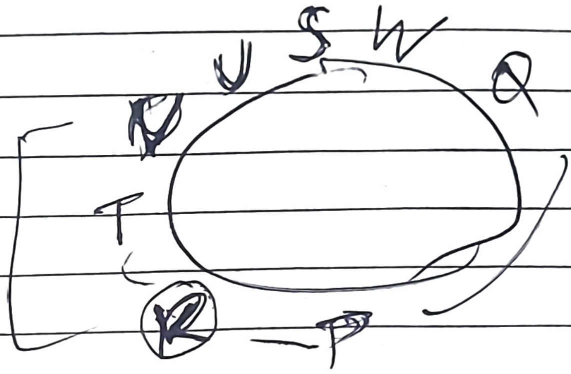
Data Arrangement :

⇒ Circular Relationship : B.

B — B A





⇒ ~~Data Arrangement~~ → A straight line arrangement

⇒ Linear Arrangement :

North

A B C D E F G

_ _ _ _ _ _ _

3 4 5 6 7

~~A B~~ C

G _ _ _ _ E F

B C D

G B C D E F A

_ _ _ _ _ _ _

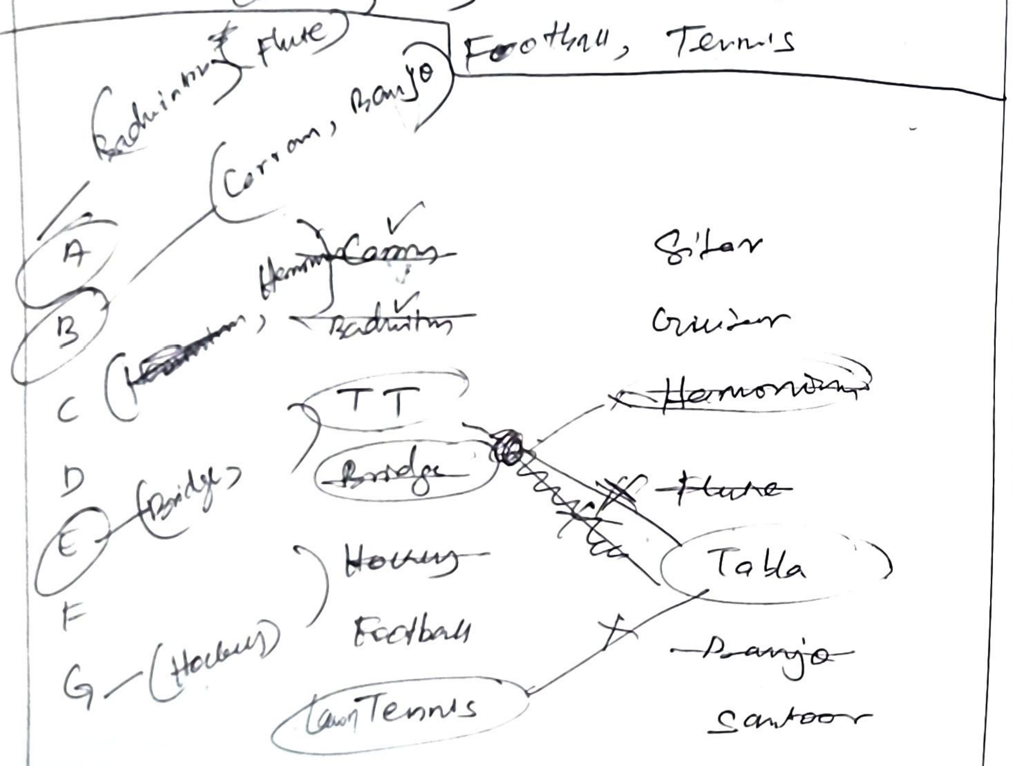
Teacher's Signature :

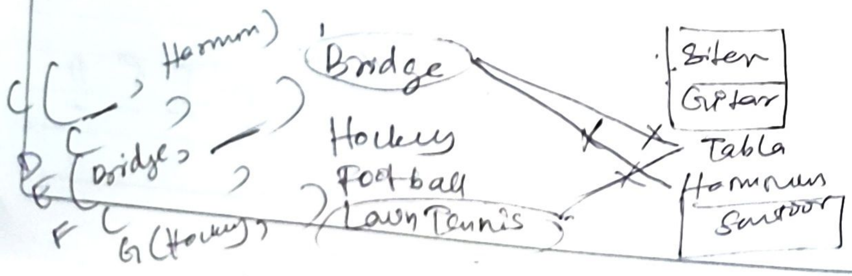
⇒ Data Arrangement Multidimensional :

Sports club..

A B C D E F G

Carrom, T.T, Badminton, Bridge, Hockey, Football, Tennis





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Ent.	Sport	Music
A	Badminton	Flute
B	Carrom	Banjo
C		Hermonium
D		Tabla
E	Bridge	~~Santoor~~
F	~~Carrom~~ B/TT	Guitar
G	Hockey	& Sitar

Instruments: ~~Flute~~, ~~Banjo~~, ~~Hermonium~~, Tabla, Santoor, ~~Sitar~~, ~~Guitar~~.

Sport: ~~BM~~, ~~Carrom~~, TT, Bridge, ~~Hockey~~, Football, LT

7. ⇒ Pipes & Cisterns :

Pipe - A : x hours if $y > x$,

Pipe - B : y hours

$$\text{net hours} = \frac{1}{x} - \frac{1}{y}.$$

Similarly

if $x > y$

$$= \frac{1}{y} - \frac{1}{x}.$$

Teacher's Signature :

$y > x$

$\frac{1}{x} - \frac{1}{y}$

Q. $A \rightarrow 12$ hours

$B \rightarrow 16$ hours

total drill $= \frac{1}{12} - \frac{1}{16}$

$= \frac{16-12}{12(16)} = \frac{\cancel{4}^{1}}{\cancel{12}_{3}(\cancel{16})16}$

Q. → Speed Time Distance.

Avg. speed $=$ $\frac{\text{Total distance travelled}}{\text{Total Time Taken}}$

Q. $\frac{1}{3}$, 20km/h

$\frac{1}{4} = 30$km

Q. Village — 25km/hr — Post office

4km/h

Of whole Journey $=$ 5 hr 48 min.

Find Distance of Village to Post office

$5 \times 60 = 300$

$+ 48$ min

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→ Village to Post office.

Distance $=$ Speed $\times$ time.

$\boxed{\text{Avg speed} = \frac{2xy}{x+y}}$

9. Relative speed Trains, Races.

40km/hr

Relative speed $= 40 - 20$

$= 20$ km/h.

$= 20 \times \frac{5}{18}$

$= \frac{\cancel{20}^{10} \times 5}{\cancel{18}_{9}}$

$= \frac{50}{9}$ m/s

20km/hr.

man

5 Sec.

$d = s \times time$

$d = \cancel{40 \times 50.5}$

$d = 40000 \times 0.5$

length of faster train $= \frac{50}{9} \times 5$ sec $= \frac{250}{9}$ sec.

$8\overline{)250}$

24

10

10. Division of wages;

Q. $A + B + C = 550$.

$\left(\frac{7}{11}\right) + (C) = \frac{9}{11}(550)$

Teacher's Signature :

(11) Mensuration:

→ Square — $A = l^2$, Perimeter $= 4L$.

→ Rectangle — Area $= L \times B$ | Perimeter $= 2(l+b)$

→ Isosceles Δ. — $A = \frac{b}{4}\sqrt{4a^2 - b^2}$, $P = 2a + b$.

→ $A = \frac{1}{2}bh$. | $P = AB + BC + CA$.

→ Equilateral Δ: $A = \frac{\sqrt{3}}{4}a^2$. | $P = 3a$.

→ Parallelogram — $A = bh$. base × height

→ Rhombus. — $A = \frac{1}{2}d_1 d_2$, $P = 4a$.

(12) Clocks:

~~① How much minute hand move per hour~~

Minute hand (360°) = Every 60 mins

Hour Hand (360°) = Every 12 hours.

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Hour hand	Minute hand
per hour how many degrees it moves: $\frac{360}{12} = 30°$	Per minute how many degree it moves. $= \frac{360}{60} = 6°$.

① Find angle b/w min hand & hour hand of a clock when time is 6:30

Time = 6:30

~~Hour hand: $6 + \frac{30}{60} = \frac{39}{6}$ hour.~~

~~360 → Every 12 hours.~~

~~x → for $\frac{39}{6}$ hour~~

~~$x = \frac{360}{12}\left(\frac{39}{6}\right)$~~

~~$x =$~~

~~$\frac{39 \times 60}{00}$, 234, 2340~~

Hour hand: $\frac{360}{12} = 30°$

Hour hand moves 0.5° per mins

At 6:30 upm.

$= 6 \times 30° + 30 \times 0.5$

$= \boxed{195°}$

Teacher's Signature:

Minute hand :

Every 60 min $= 360^\circ$

$1 \text{ min} = \frac{360}{60} = 6$

$\therefore 30\text{ min} = 30(6)$

$= \boxed{180^\circ}$

$\boxed{1 \text{ min} = 6^\circ}$

Now Angle b/w Hour hand & Minute hand $= 195^\circ - 180 = 15^\circ$

(12) Simplification:

$\boxed{\frac{x}{y} = \frac{6}{5}}$; find value of $\left(\frac{x^2+y^2}{x^2-y^2}\right)$

Sol

$$\frac{x^2+y^2}{x^2-y^2} = \frac{\frac{x^2+y^2}{y^2}}{\frac{x^2-y^2}{y^2}} = \left(\frac{\frac{x^2}{y^2}+1}{\frac{x^2}{y^2}-1}\right)$$

$$= \left[\frac{\left(\frac{x}{y}\right)^2+1}{\left(\frac{x}{y}\right)^2-1}\right] = \frac{\left(\frac{36}{25}\right)+1}{\frac{36}{25}-1}$$

$$= \frac{\frac{36+25}{25}}{\frac{36-25}{25}} = \boxed{\frac{61}{11}}$$

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Find value of x if $\frac{2x}{1+\left(\frac{1}{1+\left(\frac{x}{1-x}\right)}\right)} = 1$

Sol

$\frac{x}{1-x} \rightarrow x = 1$

$\left[\frac{(1-x)+x}{1-x}\right] = \left(\frac{1}{1-x}\right)$

$\Rightarrow \frac{2x}{1+\left[\frac{1}{\left(\frac{1}{1-x}\right)}\right]} = 1$

$\Rightarrow \frac{2x}{1+1\left(\frac{1}{\frac{1}{1-x}}\right)}$

$\Rightarrow \frac{2x}{\left(\frac{1}{1-x}\right)+1}$

$\frac{a}{\frac{b}{c}} = \frac{a}{b} \times \frac{1}{c}$

Let $a = \frac{1}{1-x}$

$\frac{2x}{1+\frac{1}{a}} = 1$

$\frac{2x}{\frac{a+1}{a}} = 1$

$\frac{2x}{a+1} \times \frac{1}{a} = 1$

$\frac{2x}{a+1} = a(a+1)$

Teacher's Signature :

$$\Rightarrow \frac{2x}{1+\frac{1}{\left(\frac{1}{1-x}\right)}} = 1$$

$$\Rightarrow \frac{2x}{1+\frac{1}{(1-x)}} = 1$$

$$\Rightarrow \frac{2x}{\frac{1-x+1}{1-x}} = 1$$

$$\Rightarrow \frac{2x}{\frac{2-x}{1-x}} = 1 \quad \Rightarrow \frac{2x}{2-x} = (1-x)$$

$$\Rightarrow 2x = 2(1-x) - x(1-x)$$

$$\Rightarrow 2x = 2-2x-x+x^2$$

$$\Rightarrow x^2-3x-2x-2 = 0$$

$$x^2-5x-2=0$$

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① Find value of $\frac{2x}{1+\frac{1}{1+\frac{x}{1-x}}}$

Sol

$$\frac{1-x+x}{1-x} = \frac{1}{1-x}$$

$$\Rightarrow \frac{2x}{1+\left(\frac{1-x}{1}\right)} = 1 \Rightarrow \frac{2x}{2-x} = 1$$

$$\Rightarrow 2x = 2-x$$

$$3x = 2 \quad \boxed{x = 2/3}$$

② If $\frac{a}{b} = \frac{3}{4}$ & $8a+5b=22$ Find val of a.

Sol Given $8a+5b=22$ — ①

÷ by b on BS

$$\frac{8a}{b} + \frac{5\cancel{b}}{\cancel{b}} = \frac{22}{b}$$

$$2\cancel{8}\left(\frac{3}{\cancel{4}}\right)+5 = \frac{22}{b}$$

$$11b = 22 \quad 2$$

$$\boxed{b=2}$$

Solve Eq ①:

$$8a + 5(2) = 22$$

$$8a = 22-10$$

$$\cancel{8}a = \cancel{12}$$

Teacher's Signature :

$$\boxed{a = 3/2}$$

Q. $999\frac{998}{999} \times 999$.

sol.

$= \left(\frac{999 \times 999 + 998}{999}\right) \times 999$

$= \left(\cancel{\frac{999\times999}{999}} + \cancel{\frac{998}{999}}\right)$

$= 999 \times 999 + (998) + 1 - 1$

$= (999 \times 999) + (999) - 1$

$= 999(999+1) - 1$

$= 999(1000) - 1$

$= 999000 - 1 = 998999$

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$64\times2 = 128$

(14) Multiplication :

$\Rightarrow (88)^2$:-

$(8)^2$	$8\times8\times2$	$(8)^2$
6400	1280	64

6400
1280
64
7744

88×88

704
704
7744

$\cancel{76\times76}$

$(76)^2$

49	$7\times6\times2$ → 84	36
00	0	

⇒ 4900 + 840 + 36 = 5776

$\Rightarrow (126)^2$

$(12)^2$	$12\times6\times2$	36
00	0	

72×2 = 144

14400
1440
36
15876

14400 + 1440 + 36

= 15876.

Teacher's Signature : ______

(15) % of increase or decrease

$$\%\ \text{change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

1) 50 to 34

$= \frac{34-50}{50} \times 100$

$= -32\%$ Means decrease.

ii) Find % increase or decrease (10 to 22)

$\frac{(22-10)}{10} \times 100$

$= 12 \times 10 = \boxed{+120\%}$

increase %.

(16) Profit / Loss

SP → Selling Price
CP → Cost Price

Profit = SP − CP
Loss = CP − SP

$\text{profit}\ \% = \frac{P}{CP} \times 100$

$\text{Loss}\ \% = \frac{L}{CP} \times 100$

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If what then what ? * * *

* If Given a selling price and its profit or loss percentage and any one asking to find cost price => then

$$CP = \frac{SP}{(\text{The percentage} \atop \text{here} -)}$$

(1) → A man incurs 10% loss by selling a bag @ 540 Rs (?) At what price should the bag be sold to earn 20% profit

Sol

Loss % = 10% => 90% of CP

SP = 540.

$CP = \frac{SP}{\text{Loss}\%} = \frac{540}{\frac{10}{100}}$

$= \frac{540}{\left(\frac{90}{100}\right)} = \frac{540 \times 10}{9} = ₹\boxed{600}\ CP$

10% Loss

Teacher's Signature :

(i) To find price to be sold to earn 20% profit.

i.e., To find SP to get 20% profit

20% profit → 120%

$$CP = \frac{SP}{\left(\frac{120}{100}\right)} \Rightarrow$$ we got it earlier

$$\boxed{CP = ₹.600}$$

~~$\frac{600}{100} \times \frac{100}{120} = SP$~~

~~₹500 = SP~~

$$\frac{600}{100} \times \frac{100}{120} = SP$$

$$\boxed{SP = ₹.500}$$

---

② CP → Shoe (bought) → SP (sold) → ₹330 profit = 10%

If it was sold at ₹.275 in Find % of loss or profit ?

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Sol

$SP = ₹.330$

Profit = 10% ⇒ $\frac{110}{100}$

$$CP = \frac{SP}{(\%)} = \frac{330}{\frac{110}{100}} = \frac{30}{\frac{1}{10}} = 30 \times \frac{1}{\frac{1}{10}}$$

$$\boxed{CP = 300\ Rs}$$

Given that If sold at ₹.275

To find % of loss or profit

~~$SP = ₹.275$~~

~~$CP = \frac{SP}{(\%)} \Rightarrow CP = \frac{SP}{x} \rightarrow$ % of increase or decrease~~

~~$\frac{x}{SP} = \frac{1}{CP} \Rightarrow \boxed{x = \frac{SP}{CP}}$~~

~~$x = \frac{275}{300} = \frac{55}{60} = \frac{11}{12}$~~

~~$\boxed{x = \frac{11}{12}\ \%}$ ⇒ Increase.~~

Teacher's Signature :

Given that CP = 300 Rs & SP = 275 Rs
it would definitely be loss. as

$CP > SP$
$300 > 275 \Rightarrow$ Loss!

To find Loss %

$\text{Loss \%} = \frac{L}{CP} \times 100$ (L → 25)

$= \frac{25}{3\cancel{00}} \times 1\cancel{00} = \boxed{\frac{25}{3} \%}$

Loss %.

(17) Basic percentages :

i) what is 33.33 % of 180.

$\frac{1}{\cancel{3}} (\cancel{180})^{6} = 180.$

$\frac{33.33}{\cancel{100}_{5}} \times \cancel{180}^{9}$

$= 33.33 \times 1.8$

$= \frac{33}{\cancel{100}_{5}} \times \cancel{180}^{9}$

$5 \overline{) 9} = 1.8$
5
40
40
0

33×1.8
264
33
59.4

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$\frac{100}{7}$. $\frac{1}{7}$ $7 \overline{) 10}$ 0.7

Tricks :

$\frac{1}{3} = 33.33\%$	$\frac{1}{6} = 16.66\%$
$\frac{1}{4} = 25\%$	$\frac{1}{7} = 14.28\%$
$\frac{1}{5} = 20\%$	$\frac{1}{8} = 12.5\%$
	$\frac{1}{9} = 11.11\%$

⇒ x is ↑/↓ by certain %.
(increased / decreased)

(P) Given $x = 200$ & it is ↑sed by 20%
Find ~~value of~~ new value of x ?

Sol

$\cancel{200}_{40} \times \frac{\cancel{120}^{6}}{\cancel{100}_{\cancel{5}}} = \boxed{\cancel{1200}\ 240.}$

(P) Kanishk salary → 20 % more than Ashok
what % Ashok salary is ~~E~~ less than kanishk

Salaries
Kanishk → x
Ashok → y.

Kanishk's salary $= y + \frac{20}{100} \Rightarrow y + \frac{1}{5}$
$= \frac{5y+1}{5}$

Teacher's Signature : ____________

18 Weighted Averages.

Averages

For consecutive integers:

→ 3 4 5 6 7

Avg = 5

→ 1 3 5 $\boxed{}$ 7 9 11

Avg = 6

Can write directly.

→ Weighted Averages

$$WA = \frac{\Sigma(\text{value} \times \text{weight})}{\Sigma(\text{weight})}$$